

FROM STATISTICS TO GENAI: COMPARING SMOTE, CTGANS AND LLMs

A focused study on Synthetic data generation for imbalanced data

ABSTRACT

Imbalanced datasets remain a persistent challenge in machine learning, such datasets when trained on will give excellent accuracy due to True negatives being in much greater in the numbers in comparison to other False Negatives and True Positives.

To improve upon such results this report dives in to different synthetic generation techniques. Using SMOTE, CTGAN, LLMs for synthetic data generation to balance the imbalanced datasets using different research papers as a base.

FOUNDATIONAL INTRODUCTION

The different techniques in this report:

1. SMOTE – Synthetic Minority Oversampling Technique, is the foundational and first major synthetic data generation technique which uses the formula of:

$$x_{new} = x_i + \lambda(x_{nn} - x_i)$$

- x_i = original minority sample
- x_{nn} = Random nearest neighbour
- $\lambda \in [0,1]$

This method ensures that the new samples are not exact copies but interpolations in the dataset.

2. CTGAN – Conditional Tabular Generative Adversarial Network, is the newer approach to the synthetic data generation, published in 2019. This approach uses two deep learning models competing each other:
 - Generator – Generating new synthetic data points from the dataset, which get mixed with the real data.
 - Discriminator – Identifies whether the data point is real or synthetic.

These two models compete with each other to give the best result in complex set of data.

3. LLMs – Large Language Models, although these are based on transformers and are used to generate human-like language, but since these are trained

METHOD FLOW

SMOTE:

Original Dataset → Identify Minority Class → Select Nearest Neighbours → Interpolation → Synthetic Samples

CTGAN:

Tabular Data → Generator + Discriminator Training → Learn Data Distribution → Generate Synthetic Data

LLMs:

Tabular Data → Convert to Contextual/Text Format → Prompt LLM → Generate Semantic Data → Convert back to Tabular

on large amount of data these can be easily used to generate new semantic data once the tabular data is contextualized.

COMPARATIVE ANALYSIS

METHODS	SMOTE	CTGAN	LLMs
Data Generation Mechanism	This method generates samples linearly, hence assuming the feature space to be linear.	This method uses the training to learn the joint distribution of tabular data. Hence it can model both numerical and categorical features.	This method generates new data by leveraging context and the semantic relationships between features without the requirement of probabilistic model of dataset
Statistical Fidelity	This method struggles in maintaining global distribution structure as it only generates local neighbourhoods	CTGANs generally produce higher fidelity data, and preserves categorical distributions and correlations	LLM based methods like GReaT preserve the correlations and semantic consistency outperforming CTGAN in statistical properties.
Classification Performance	SMOTE achieved best performance with high recall and F1 score for minority class detection	Improved performance over the baseline but did not outperform SMOTE due to data and tuning requirements.	Show promising results but are less consistent for downstream predictive tasks
Ability to Capture Complex Relationships	Cannot capture non linear or global relationships as it only interpolates between nearby points	Can model complex, non-linear dependencies, but requires large datasets and careful tuning	These excel at logical and contextual relationships between features as they model data as sequences and learn dependencies implicitly
Data Requirements and Practical Usability	Works well even with small datasets and requires minimal computation.	Requires large datasets (>10,000 samples) and significant hyperparameter tuning to perform well	Can work in low-data or even zero-shot settings, but may require prompt engineering or formatting strategy
Limitations	Can't handle categorical data well and may create overlapping classes	Training instability, High computational cost with sensitive hyperparameter tuning	Difficulty in maintaining strict numerical constraints and may generate logically inconsistent rows without guidance

CONCLUSION

SMOTE technique remains strong baseline for the data generation due to its simplicity and effectiveness in improving classification especially in low data scenarios, but when it comes to a more sophisticated non-linear dataset CTGAN provides high effectiveness but depends heavily on data size and tuning.

LLMs on the other hand represent a new standard offering superior flexibility and ability to capture semantic and logical relationships, though they still require refinement for structured data consistency.

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